Repairing the Negation Blind Spot in Hebrew Text Embeddings

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Submitted as final project report for the NLP course, Reichman University, 2026

Abstract

Sentence embedding models are widely assumed to be sensitive to meaning, but negation is a direct test of that assumption: a sentence and its negation share nearly every surface word while meaning the opposite thing. We measure this *negation blind spot* in four frozen Hebrew-capable embedders using a hand-verified probe mined from HebNLI (303 triples), and find it real: two of four models score a sentence's negation as more similar to it than a faithful paraphrase. We test two post-hoc repairs. `projection` amplifies each sentence's own component along an estimated negation direction in the frozen embedding space; naively selecting its amplification strength by the headline metric alone collapses the representation (unrelated sentences become near-identical, sim_unrelated=0.975) rather than fixing it, and only a second, independent guard metric catches this. `nli_rerank` instead blends the frozen cosine with a decontaminated, task-specific Hebrew NLI classifier under a principled selection rule for the blend weight, reaching $\geq$0.986 pairwise accuracy on every model. The paper's contribution is less the two fixes themselves than the shared methodological finding behind them: gap-only selection of a representation-editing strength is unsafe by construction, cross-validation does not protect against it, and a real trade-off guard is necessary rather than optional.

1 Introduction

Sentence embedding models are trained to place sentences with similar meaning close together in vector space, and are used downstream as a drop-in similarity function for retrieval, clustering, and semantic search. That use carries an implicit assumption: that the embedding space is sensitive to truth-conditional meaning, not just to topic. Negation is the sharpest test of that assumption. A sentence and its negation share almost every surface word, cover the identical topic, and yet mean the opposite thing. If an embedding model is really encoding meaning, cosine similarity should treat a paraphrase as close and a negation as far. If the model is largely encoding lexical overlap, it treats a sentence and its negation as near-duplicates — the *negation blind spot*.

This is not a hypothetical failure mode. Weller et al. [2024] show that neural retrieval models, including strong bi-encoders, rank documents differing only by negation close to chance. Ravichander et al. [2022] show the same gap in reading comprehension: state-of-the-art QA models answer questions about a passage and its negated counterpart inconsistently far more often than humans do. Concurrent work diagnoses the same problem directly in sentence embedding spaces [Cao, 2025], confirming it is a property of the representation, not just of a particular downstream task.

This project asks whether the same blind spot exists in Hebrew sentence embeddings, and whether it can be repaired without retraining the underlying model. Hebrew is a useful and under-studied test case: negation is carried by a small, closed set of dedicated particles (לא, אין/אינו, ללא/בלי/בלתי, quantifier phrases like אף אחד and אף פעם) that we stratify into six grammatical types (Section 2.1), and no prior work we are aware of has measured or addressed the negation blind spot specifically for Hebrew embeddings.

Measuring the blind spot

We define the **cosine gap** for a (target, paraphrase, negation) triple as gap $= \cos(target, paraphrase) - \cos(target, negation)$. A model that is sensitive to negation should have a gap well above zero. We measured this on four frozen, publicly available embedding models usable for Hebrew — `multilingual-e5-base`, `LaBSE`, `sentence-transformers-alephbert`, and `sambert` — on our held-out test split (151 triples, Section 2.1):

Two of the four models place the negation marginally *closer* to the target than the paraphrase. By any of these models' own metric, a sentence and its opposite are nearly indistinguishable from a sentence and a faithful restatement of it.

model	cos(target, paraphrase)	cos(target, negation)	cosine gap
multilingual-e5	0.963	0.941	**+0.022**
LaBSE	0.890	0.824	**+0.066**
alephbert-sentence	0.871	0.890	**−0.020**
sambert	0.879	0.896	**−0.017**

Table 1: Baseline cosine gap, four frozen embedders, 151-item test split.

Research questions

1. Can the blind spot be repaired post-hoc, without fine-tuning the embedding model itself, by intervening on the vector space directly?
2. How does a representation-space fix compare to a signal-fusion fix — blending the frozen embedding's cosine with an external Hebrew NLI classifier's judgment — on the same probe and the same models?
3. Does a fix that looks successful on its own headline metric actually work, or can it "succeed" by degenerate means (e.g. collapsing the representation so that everything looks equally similar to everything else, which trivially opens any gap)? This turned out not to be a hypothetical concern (Section 3).

We treat (1) as the `projection` intervention, (2) as the `nli_rerank` intervention, and (3) as a methodological finding that ended up shaping both.

1.1 Related Work

Diagnosing the negation blind spot. Negation insensitivity in neural text representations has been documented across tasks and architectures. Weller et al. [2024] build a retrieval benchmark of document pairs differing only by negation and find that bi-encoder and sparse retrievers — the same family our frozen embedders belong to — perform at or below chance. Ravichander et al. [2022] show the analogous gap in reading comprehension via contrastive edits to a passage (paraphrase, scope change, polarity reversal) — the same paraphrase-vs.-negation-around-a-shared-target design our probe schema is built on. Concurrent 2025 work [Cao, 2025] diagnoses negation blindness directly inside universal sentence embedding spaces and proposes an adapter module trained on top of a frozen embedder to fix it, confirming the phenomenon is representational rather than task-specific. Their fix and ours share the frozen-base-model setup; the real difference is that their adapter is a trained module, where `projection` needs no training at all — only direction estimation and a closed-form rescaling (`nli_rerank` does require training a classifier, so this contrast is specific to `projection` and does not extend to both of our interventions).

Fixing it by editing the representation. Finding a linear "concept direction" inside an embedding space and manipulating a vector's component along it is well established, but the two established use cases point in opposite directions from what we need. Hard-debiasing [Bolukbasi et al., 2016] finds a direction correlated with an unwanted attribute and *removes* it, so the attribute stops influencing similarity. Activation steering and representation-engineering methods [Turner et al., 2023, Zou et al., 2023] instead *add* a scaled concept vector to steer a model's behavior toward a target concept at inference time. Our `projection` intervention needs neither: we are not trying to erase negation's influence (that would make a sentence and its negation *more* alike) nor add an unrelated steering concept, but amplify the sentence's own existing component along an estimated negation direction — closer in spirit to activation steering's amplification mechanism than to debiasing's removal, but applied per-sentence to a signal the sentence already carries.

Fixing it by fusing an external signal. An orthogonal approach is to leave the embedding alone and combine its score with a second, negation-aware signal at scoring time — the strategy behind cross-encoder re-ranking in NevIR and behind our `nli_rerank` intervention, which blends frozen cosine similarity with a Hebrew NLI model's entailment/contradiction judgment.

Hebrew resources. We build on AlephBERT [Seker et al., 2022], the Hebrew pre-trained language model underlying both the `alephbert-sentence` embedder and the NLI classifier fine-tuned for `nli_rerank`, and on the HebNLI dataset [HebArabNlpProject] as the source corpus our negation probe is mined from.

What is missing that this project addresses. To our knowledge, the negation blind spot has not previously been measured or repaired for Hebrew sentence embeddings specifically, and prior post-hoc representation edits in this space are typically reported against a single headline metric without a guard against the intervention degrading the representation elsewhere. We found that guard to be necessary rather than optional (Section 3).

2 Methodology

2.1 Dataset construction

No existing Hebrew negation probe fit this project's needs, so we built one from HebNLI's `train` split, the Hebrew counterpart of MultiNLI. Each usable item is a triple: a **target** sentence, a **paraphrase** that preserves its meaning, and a **negation** that reverses it through a negation marker and nothing else. HebNLI's own `contradiction` label is not a proxy for this — a contradiction can equally arise from an antonym, a changed number, or a world-knowledge fact, none of which involve negation at all.

An automatic filter narrows HebNLI's 100,390 train prompts to candidates before any human review:

stage	pairs remaining	share of previous stage
HebNLI prompts	100,390	—
has a contradiction pair	96,929	97%
well-formed length	86,352	89%
negation marker was added	32,733	38%
minimal single-marker edit	689	2%

Table 2: Mining funnel. 62% of HebNLI's contradiction pairs contain no negation marker at all.

Each of the 689 surviving candidates was reviewed by hand against a fixed question: is the opposition between target and negation carried by a negation marker, and by nothing else? **303 of 689 (44%) were accepted.** The probe is split 152 (train) / 151 (test), deterministically and stratified by negation type so no type ends up entirely on one side:

type	example marker(s)	share of 303
particle	לא	58%
quantifier	שום, אף פעם, אף אחד	18%
existential	אינם/אינה/אינו, אין	16%
question	negation inside an interrogative	4%
privative	אי־, בלתי, בלי, ללא	2%
neg-raising	matrix-clause negation of a raising predicate	2%

Table 3: Negation-type stratification of the 303-item probe.

Inter-annotator agreement was measured via Cohen's kappa on an independent 40-item random sample double-annotated by both authors against a fixed guideline document: raw agreement 77.5% (31/40), $\kappa = 0.543$ (moderate). Of the 9 disagreements, most turned on how strictly to read "differs only in polarity" — one author accepted a few candidates with a minor synonym-level verb change alongside the negation marker, the other rejected any lexical drift beyond the marker itself; a smaller cluster concerned whether a title-like fragment with no verb counts as a "proposition" at all.

Because the probe is mined from HebNLI's own train split, any NLI model later fine-tuned on HebNLI for `nli_rerank` risks having already seen the probe's (target, negation) pairs with their gold `contradiction` label. All 689 mined promptIDs are held out of NLI fine-tuning data by promptID (not pairID, since MultiNLI gives three hypotheses per premise sharing one promptID).

2.2 Harness

For every (embedder, intervention) pair we compute, on the held-out test split: cosine gap (defined above), pairwise accuracy — the share of items where score(*target*, *paraphrase*) > score(*target*, *negation*), chance = 0.5 — and, where wired, Pearson/Spearman correlation against a Hebrew Semantic Textual Similarity set as a trade-off guard.

2.3 Hebrew STS-B translation

The English STS Benchmark aggregates sentence pairs from the SemEval Semantic Textual Similarity shared tasks run between 2012 and 2017 [Agirre et al., 2017]. We translate its official `dev` (1,500 pairs) and `test` (1,379 pairs) splits into Hebrew, keeping the original pair IDs, splits, and gold similarity scores untouched; the 5,749 training pairs are not translated, since no model here is trained or tuned on STS-B itself. Each pair goes through two independent LLM translators with no visibility into each other's output, then a third LLM agent adjudicates — keeping the stronger candidate or merging the two into a corrected Hebrew pair when neither is adequate. Gold scores are attached only after adjudication and automated validation are complete, and are never adjusted to fit a translation. A human reviewer additionally inspected roughly 200 translated items for correctness and preservation of meaning, and found them satisfactory. Machine-translated STS benchmarks are known to introduce translation artifacts [Isbister and Sahlgren, 2020]; we treat the inherited English gold scores as a practical cross-lingual approximation rather than assume Hebrew speakers would assign identical scores in every case (see Limitations).

2.4 Intervention 1: `projection` (representation-space edit)

Estimates a negation direction d from the train split — either the mean difference between negation and target embeddings, or the normal to a logistic-regression hyperplane separating {target, paraphrase} from {negation} — and rescales each vector's own component along d: $v' = v + (\gamma - 1) \cdot ((v \cdot d) - \mu) \cdot d$. $\gamma = 1$ is the identity; $\gamma > 1$ amplifies the sentence's existing signal along the negation direction, deliberately not the debiasing move of projecting the direction out.

γ is selected by sweeping a grid and maximizing the train-split cosine gap under 5-fold cross-validation, never touching test. The naive version of this — argmax over gap alone — turned out to be unsafe (Section 3: it collapses the representation). The fix constrains selection to γ values whose cross-validated sim_unrelated (mean $|\cos|$ between the targets of two unrelated probe items) stays at or below a fixed threshold (0.5), then argmaxes the gap among only those. This selects γ within one fixed (direction estimator, centering, CV-vs-train) configuration; we separately ablate all 8 such configurations per model (direction $\in$ {mean-diff, classifier} $\times$ centering $\in$ {center, nocenter} $\times$ selection $\in$ {cv, train}) as a sensitivity check (`results/projection_ablation.csv`), but do not headline the best of the 8 per model: each row's accuracy is itself a test-split number, so picking per model whichever scores highest would be test-set selection across 8 candidates. We report one configuration instead, fixed in advance and applied identically to every model: mean-diff direction, centered, γ by cross-validation — `NegationProjection`'s own defaults.

2.5 Intervention 2: `nli_rerank` (signal-fusion)

Blends the frozen embedder's cosine with a directional Hebrew NLI classifier's judgment: $\text{nli\_score}(a, b) = P(\text{ent} \mid a, b) - P(\text{contra} \mid a, b)$, $\text{combined}(a, b, \lambda) = (1 - \lambda) \cdot \cos(a, b) + \lambda \cdot \text{nli\_score}(a, b)$. $\lambda = 0$ is pure cosine (the baseline); $\lambda = 1$ discards the embedder entirely.

Classifier. AlephBERT fine-tuned for 3-way NLI on HebNLI (lr $= 2 \times 10^{-5}$, batch size 32, 2 epochs, sequence length 128). Test accuracy 79.6% (macro-F1 79.4%) on the clean HebNLI test split (883 pairs) — see Results for how this compares to the ready-made alternative, `oriel9p/AlephBERT-FT-HebNLI-LCHAIM`. Checkpoint: `https://huggingface.co/CodingBz/alephbert-hebnli-clean`. **Why not the released checkpoint.** That checkpoint is AlephBERT fine-tuned on HebNLI together with LCHAIM [Malul et al., 2025], a Hebrew NLI dataset built to evaluate long-context reasoning over multi-sentence passages translated from English ConTRoL. That training objective is not aligned with our task — our probe's targets have a median of 6 tokens, not long premises — so we fine-tune our own classifier instead, which the comparison above supports. **Decontamination.** All three HebNLI splits used for fine-tuning are filtered by the 689 mined promptIDs, plus a text-level audit catching residual overlap under a different promptID (9/1/0 rows dropped from train/val/test). Final fine-tuning set: 297,990 train / 1,989 val / 883 test pairs.

Choosing λ. λ is selected per embedder on a 21-point grid (0.00–1.00, step 0.05), not by maximizing probe accuracy alone. The rule: among λ whose Hebrew-STS-dev Spearman correlation stays within 0.02 (absolute) of that same embedder's own $\lambda = 0$ Spearman, take the highest pairwise accuracy; ties broken by the largest mean gap, then the smallest λ. Selection uses the train split (152 items) and Hebrew STS-B *dev* (1,500 pairs, translated from English STS-B); the selected λ is locked and evaluated exactly once on the test split (151 items) and Hebrew STS-B *test* (1,379 pairs).

Both interventions are fit/selected on train only, and reported on test and Hebrew-STS-*test* only; `nli_rerank` additionally uses Hebrew-STS-*dev* as a selection-time guard, which `projection` does not have (it uses sim_unrelated under cross-validation on train instead — see Limitations for why that is a weaker guard).

3 Experimental Results

3.1 `projection`: gap-only selection is unsafe

The unconstrained version of γ-selection — maximize the train-split gap and nothing else — always won by amplifying until the representation collapsed. Representative case, `sambert`, mean-difference direction, cross-validated selection:

selection	γ	cosine gap	pairwise acc.	sim_unrelated
unconstrained	1000	+0.899	0.89	**0.975**
constrained (≤ 0.5)	12	+0.338	0.88	**0.350**

Table 4: At $\gamma = 1000$, sim_unrelated is 0.975 — the representation has collapsed onto the negation direction.

This pattern held across every model and direction estimator tried: unconstrained γ landed at the top of the search grid every time, with sim_unrelated in the 0.94–1.00 range. Constraining selection fixes this on three of four models; **`multilingual-e5` is the exception** — every constrained configuration still fails to bring sim_unrelated under 0.5 anywhere in the grid, falling back to the grid's lowest-unrel point (0.59–0.79). We report this as a negative result specific to this embedder.

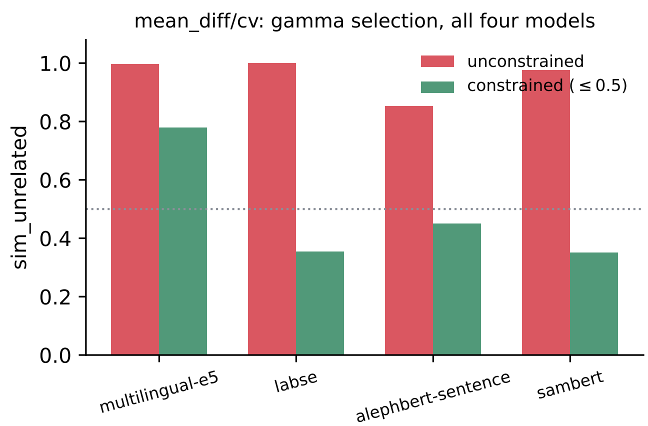

Figure 1: Same phenomenon, all four models, mean-difference direction, cross-validated selection: unconstrained selection lands on sim_unrelated $\geq$ 0.85 every time; the constrained version lands under 0.45 on three of four (multilingual-e5 is the exception discussed above).

3.2 `nli_rerank`: locked test-set numbers

Our fine-tuned classifier reaches 79.6% test accuracy (macro-F1 79.4%) on the clean HebNLI test split, against 72.7% (macro-F1 72.2%) for the released `LCHAIM` checkpoint on the same split. This is not a fair comparison, but not for the reason it might first appear: it is not fair *in the classifier's own favor*, since the released checkpoint was fine-tuned on all of HebNLI rather than a held-out split, so this test set was very likely part of its own training data — a documented caveat on that evaluation run. Contamination should inflate a model's score, not deflate it, so the released checkpoint scoring seven points lower despite that likely advantage is stronger evidence for training a task-specific classifier than a same-objective comparison would have given us. The same pattern holds directly on the probe task itself, not just generic HebNLI: scoring both checkpoints with $\lambda = 1$ (pure NLI, no cosine contribution) on `multilingual-e5`'s probe pairs, our classifier reaches 0.993 pairwise accuracy against LCHAIM's 0.987. Both classifiers' weakest class is *neutral*, but by different margins — our classifier's neutral recall is 0.740, LCHAIM's is 0.575, while LCHAIM over-predicts *contradiction* in its place.

model	λ	pairwise acc. (λ=0 → selected)	STS Spearman (λ=0 → selected)	test drop
multilingual-e5	0.05	0.80 → **0.99**	0.76 → 0.76	0.007
LaBSE	0.35	0.79 → **0.99**	0.74 → 0.67	**0.070**
alephbert-sentence	0.30	0.50 → **0.99**	0.65 → 0.65	0.000
sambert	0.35	0.50 → **0.99**	0.69 → 0.67	0.014

Table 5: Every model reaches ≥0.986 pairwise accuracy. The 0.02 STS-Spearman-drop budget is a *dev-time* selection criterion (see Methodology); the rightmost column is what that drop actually turned out to be on the locked *test* split. Three of four models stay well inside the dev-time budget (test drop ≤0.014). LaBSE does not: its dev-time margin was already narrow (0.019 of the 0.02 budget), and that margin did not survive contact with the held-out split — its test-time drop is 3.5× the budget it was selected under. This is a real result about the selection rule's generalization, not a rounding error, and is discussed further in Limitations.

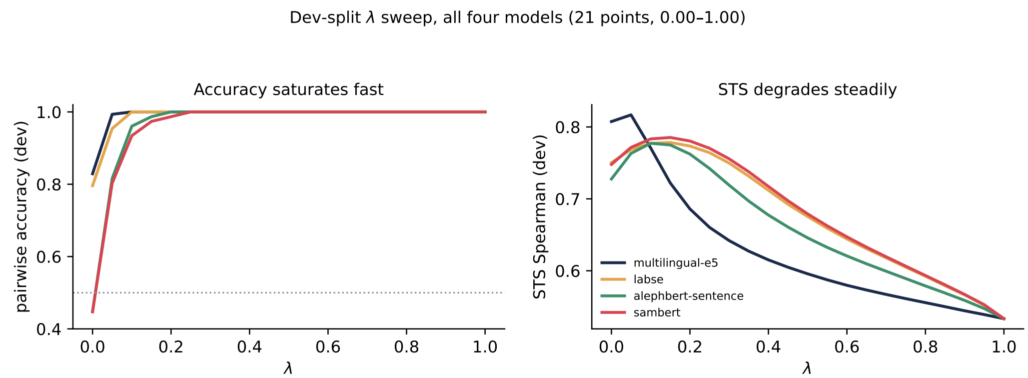

Figure 2: Dev-split λ sweep behind "a small λ is enough": pairwise accuracy saturates near $\lambda \approx 0.1$–0.2 for every model, while STS Spearman rises slightly and then declines from there — the two curves justify why the selection rule finds a small λ sufficient rather than needing to trade deep into the STS budget.

3.3 Head-to-head

3.4 Where the fixes still fail: per-category breakdown

The three largest categories — particle, existential, quantifier, together 89% of the test items — all see `projection` close roughly 75–83% of the baseline gap; `question` recovers similarly despite the lowest starting point of any category. The two categories where `projection` struggles most in relative terms are the two smallest and linguistically most specific: privative and neg-raising close only 32% and 53% of their gaps. Both are low-n enough (12 items across all four models combined) that this reads as a real but uncertain signal rather than a confident per-category claim.

4 Discussion

We confirmed the negation blind spot exists in Hebrew sentence embeddings — two of four frozen models score a sentence's negation as *more* similar to it than a faithful paraphrase — and tested two ways to repair it without retraining the embedder. `nli_rerank` closes the gap almost completely on every model (≥0.986 pairwise accuracy) while a semantic-similarity guard stays within its dev-time budget for three of four models (Results; LaBSE is the exception). `projection` recovers a substantial share of the gap without a second model at inference time, but only once its own selection procedure is constrained against a collapse guard.

The methodological finding, not just the fix, is the contribution. The unconstrained-γ failure mode fell out of widening the search grid on real models and noticing the headline metric kept improving in a direction that should have been suspicious. The contribution is that amplifying a concept direction safely requires a second, independent guard metric; that gap-only selection reliably finds the degenerate optimum instead of the intended one; and that cross-validation does not protect against this particular failure, because collapse improves the metric on every fold, held out or not. `nli_rerank`'s λ-selection rule was built on the same principle independently (a hard

model	intervention	pairwise acc.	trade-off guard
	baseline	0.80	STS 0.76
multilingual-e5	projection (mean-diff, γ=12, relaxed)	0.97	sim_unrel 0.78 (over budget)
	nli_rerank (λ=0.05)	**0.99**	STS 0.76
	baseline	0.80	STS 0.74
LaBSE	projection (mean-diff, γ=5)	**0.99**	sim_unrel 0.35
	nli_rerank (λ=0.35)	**0.99**	STS 0.67 (0.070 test drop)
	baseline	0.50	STS 0.65
alephbert-sentence	projection (mean-diff, γ=12)	0.97	sim_unrel 0.45
	nli_rerank (λ=0.30)	**0.99**	STS 0.65
	baseline	0.50	STS 0.70
sambert	projection (mean-diff, γ=12)	0.88	sim_unrel 0.35
	nli_rerank (λ=0.35)	**0.99**	STS 0.67

Table 6: `projection`'s own default configuration (mean-difference direction, centered, cross-validated γ), constrained to sim_unrelated $\leq$ 0.5, applied identically to all four models — not the best of the 8 direction/centering/selection combinations we separately ablate (Section 3), since picking per model whichever of those scores highest on test would itself be test-set selection (multilingual-e5: the constraint is unsatisfiable here, so the lowest-*sim_unrelated* point is shown, flagged relaxed). sim_unrelated and STS are on different scales and not directly comparable, so both guards are reported alongside rather than merged; `nli_rerank` wins on every model on accuracy, with a real semantic-similarity guard rather than a proxy, except that its own guard did not hold at test time for LaBSE (Table 5). `projection` needs no second model at inference time and closes most of the accuracy gap on three of four models, sambert being the exception.

category	n (×4 models)	baseline pass rate	projection pass rate	gap closed
particle (לא)	356	0.67	0.92	76%
existential (אין)	100	0.65	0.94	83%
quantifier	108	0.61	0.93	82%
question	16	0.25	0.81	75%
privative	12	0.75	0.83	32%
neg-raising	12	0.83	0.92	53%

Table 7: Per-category pass rate, aggregated across all four models, baseline vs. `projection`.

STS budget, not a weighted trade-off), and the two interventions validate each other on this point.

4.1 Limitations

- **Probe size.** 303 items (151 test) is small; the smallest per-category slices (3–4 items per model) are suggestive, not conclusive.
- sim_unrelated **is a proxy, not a real similarity benchmark.** It is exactly what let `nli_rerank`'s STS guard catch a generalization failure (LaBSE, Table 5) that `projection`'s proxy guard has no way to catch, since it is only measured on train via cross-validation and never checked against a held-out split at all; `projection` would benefit from the same held-out Hebrew-STS guard `nli_rerank` already has, which we did not have time to wire in.
- **Hebrew STS-B is a translation** of the English benchmark [Agirre et al., 2017] with inherited English gold scores, not natively annotated in Hebrew; machine-translated STS benchmarks are known to introduce translation artifacts [Isbister and Sahlgren, 2020], and our two-translator-plus-adjudicator pipeline with a human spot-check reduces but does not eliminate that risk.
- **The 0.02 STS budget is a dev-time guarantee, not a test-time one.** It held on test for three of four models but not LaBSE, whose dev-time margin was already narrow (Table 5) — a budget checked once on dev does not certify every model's behavior on unseen data.
- **No γ in our search grid resolves `multilingual-e5`'s** collapse guard.

- **Single-run test evaluation.** Each configuration is evaluated once on the locked test split rather than averaged over seeds, appropriate for the "no peeking" guarantee but meaning we cannot separately quantify the variance in any single reported number.

4.2 Future work

Wire a held-out Hebrew STS guard into `projection`'s γ-selection, matching `nli_rerank`'s dev/test discipline. Extend the probe with a fourth sentence per item to enable a true NevIR-style 0.25-chance metric. Investigate why `multilingual-e5` resists the collapse-safe region entirely, and why LaBSE's STS margin does not generalize from dev to test while the other three models' do.

5 Code

Full code, data, and results: `https://github.com/ItayBoros/hebrew-negation-embeddings` (branch `main`). Colab notebooks for the GPU-dependent stages (probe mining, real-model evaluation, NLI fine-tuning, λ selection) are under `notebooks/`.

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